

Extend BAT FUNCTION



Storage&Battery

Introduction

For customer who wants more batteries to extend system capacity, excessive SOC difference (more than 30%) between old & new batteries may lead problem, such as SOC sudden step or power limiting. Extend BAT FUNC is for the convenience of extending batteries to avoid problem caused by excessive SOC differences.

SOC Difference Over 30% Between Old and New Batteries Leads to Issues				
Old Battery System			+	New Battery
Battery health 90%	...	Battery health 90%		Battery health 100%
Usable Capacity 90%		Usable Capacity 90%		Usable Capacity 90%
SOC 80%		SOC 80%		SOC 45%

Extend BAT FUNC (Settings → Advance settings → Extend BAT FUNC) will force all old battery SOC within 36%~40% in next 48h meanwhile all new battery SOC around 45%~48% (The battery's SOC will decrease by approximately 1% per month).

Old Battery System			+	New Battery
Battery health 90%	...	Battery health 90%		Battery health 100%
Usable Capacity 90%		Usable Capacity 90%		Usable Capacity 90%
SOC 36%~40%		SOC 36%~40%		SOC 45%~48%

Advantages

Safe

Avoid battery expansion problem caused by SOC difference.

Simple

Enable and forget, Extend BAT FUNC will disabled by default after 48h.

Convenient

No need to bother how to control old battery SOC.

Priority

Export control > Extend BAT FUNC > work mode

Export control has higher priority to meet 0 injection. In this case Extend BAT FUNC may not able to drop SOC to 36%~40% quickly if no load & 0 injection.

Compatibility

Only G4 inverter support Extend BAT FUNC.

Note:

system usable capacity = the lowest battery usable capacity in system * batteries quantity So if old battery health only left 90% usable capacity, the new 100% health battery only able to add 90% usable capacity to system.

Old Battery System			+	New Battery	
Battery health 90%	...	Battery health 90%		Battery health 100%	10% usable capacity will be wasted
Usable Capacity 90%		Usable Capacity 90%		Usable Capacity 90%	